

Evaluating Impacts of Climate Change on GDP

Muhammad Ibrahim Farrukh

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Contents

1	Introduction	2
2	Data and Initial Exploratory Analysis	2
2.1	Data structure and coverage	2
2.2	Outcome transformation	2
3	Historical Climate–GDP Comparisons	2
3.1	Bivariate relationships	2
4	Geographic Visualizations of Historical Change	3
5	Modeling Approach and Historical Effect Estimation	4
5.1	Empirical strategy	4
5.2	Model selection and diagnostics	4
5.3	Estimated climate–GDP relationships	5
6	Projections of Future Climate Change Impacts on GDP	5
6.1	Projection methodology	5
6.2	Projected paths and cross-country heterogeneity	5
7	Interpretation and Policy Implications	6
8	Recommendations for Strengthening the Assessment	6
9	Conclusion	7

1 Introduction

In a policy context, the World Bank and related stakeholders increasingly rely on empirical estimates of climate economy relationships to design climate insurance instruments. This report addresses five objectives requested by a program officer evaluating climate impacts on GDP.

2 Data and Initial Exploratory Analysis

2.1 Data structure and coverage

The historical dataset (`Historical.csv`) is a country–year panel covering 209 countries from 1961 to 2020 (10,277 observations). For each country–year, it reports GDP per capita, average annual temperature ($^{\circ}\text{C}$), and total annual precipitation (mm). The future dataset (`Future.csv`) contains projections for six countries (China, Greece, India, Iran, Kuwait, and the United States) from 2021 to 2100 under two climate scenarios (SSP1-2.6 and SSP5-8.5), including projected temperature and precipitation. Both datasets contain no missing values, simplifying subsequent estimation and projection exercises.

2.2 Outcome transformation

GDP per capita in levels is highly right-skewed, spanning more than two orders of magnitude across countries and years. Such skewness can allow extreme observations to dominate regression results and obscure underlying relationships. Figure 1 shows the distribution of log GDP per capita, which is substantially more symmetric and approximately unimodal. Accordingly, all subsequent analysis models $\log(\text{GDP per capita})$ as the primary outcome variable.

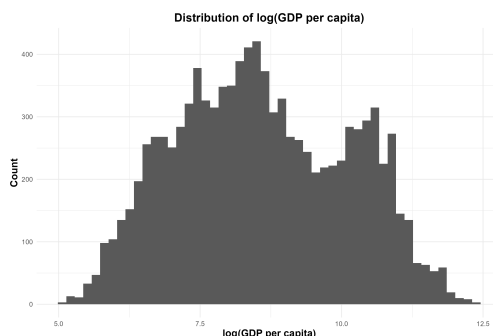


Figure 1: Distribution of log GDP per capita in the historical panel (1961–2020).

3 Historical Climate–GDP Comparisons

3.1 Bivariate relationships

Figure 2 summarizes the unconditional historical relationships between climate variables and log GDP per capita using hexbin density plots with LOESS smoothers, which reveal central tendencies despite heavy overplotting.

Figure 2a shows a clear nonlinear, inverted-U relationship between temperature and income: log GDP per capita is highest at intermediate temperatures and lower at both extremes. This visual evidence indicates that a purely linear temperature effect would be misspecified and motivates the use of flexible functional forms in the regression analysis.

By contrast, Figure 2b shows weaker and noisier association between precipitation and log GDP per capita. Although some curvature is visible, the overall relationship is less pronounced, suggesting that precipitation plays a secondary and more heterogeneous role in simple bivariate comparisons.

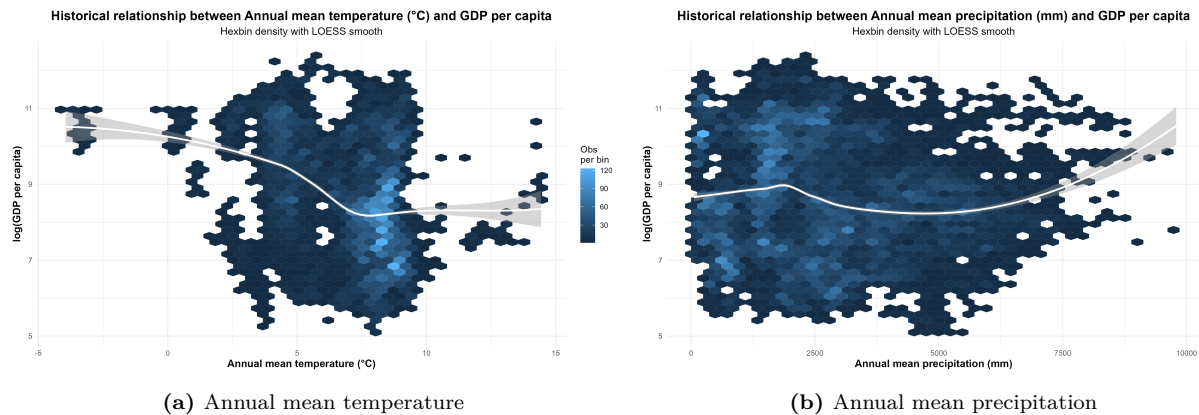


Figure 2: Historical relationship between climate variables and log GDP per capita.

4 Geographic Visualizations of Historical Change

Figures 3, 4a, and 4b map changes in country-level average log GDP per capita, temperature, and precipitation between 1981–1990 and 2011–2020. For each variable, I compute the difference in period averages (late minus early). This window is chosen because it retains broad country coverage (190 of 209 countries) while providing sufficient temporal separation to capture long-run economic and climatic change.

Figure 3 reveals substantial heterogeneity in long-run growth, with large gains in East and Southeast Asia and more modest or stagnant outcomes in parts of Sub-Saharan Africa and the Middle East. Figure 4a shows widespread warming across most regions, especially in the Northern Hemisphere, while Figure 4b exhibits much more spatially uneven changes, with both increases and decreases across regions.

Overall, these maps illustrate that both economic and climatic changes are highly uneven across countries, motivating the use of country fixed effects and an identification strategy based on within-country variation over time.

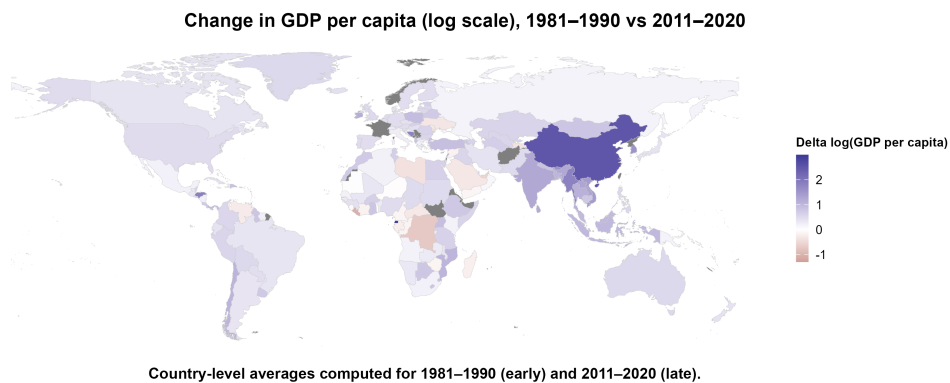


Figure 3: Change in average log GDP per capita between 1981–1990 and 2011–2020. Values represent country-level differences in period averages (late minus early).

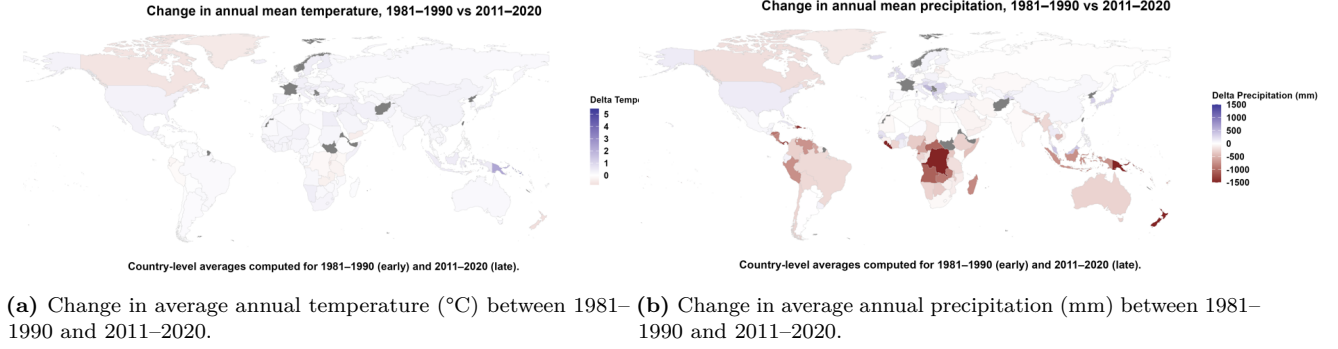


Figure 4: Changes in country-level average climate conditions between 1981–1990 and 2011–2020. Values represent differences in period averages (late minus early).

5 Modeling Approach and Historical Effect Estimation

5.1 Empirical strategy

I model log GDP per capita as a function of temperature and precipitation using increasingly flexible specifications, with country and year fixed effects to ensure identification comes from within-country variation over time.

Let i index countries and t years. The baseline parametric specification is a quadratic fixed-effects model:

$$\log(\text{GDP}_{it}) = \beta_1 T_{it} + \beta_2 T_{it}^2 + \beta_3 P_{it} + \beta_4 P_{it}^2 + \alpha_i + \gamma_t + \varepsilon_{it}, \quad (1)$$

where T_{it} is average temperature, P_{it} is precipitation, α_i are country fixed effects, and γ_t are year fixed effects. Standard errors are clustered at the country level.

To allow for more flexible nonlinear responses, I also estimate a generalized additive model (GAM):

$$\log(\text{GDP}_{it}) = f_1(T_{it}) + f_2(P_{it}) + \alpha_i + \gamma_t + \varepsilon_{it}, \quad (2)$$

where $f_1(\cdot)$ and $f_2(\cdot)$ are smooth functions estimated using penalized regression splines.

5.2 Model selection and diagnostics

I estimate three specifications: a quadratic fixed-effects model, a GAM with $k = 10$, and a GAM with $k = 20$, where k controls spline flexibility. Basis-dimension diagnostics report a k -index with an associated p-value, where low p-values and effective degrees of freedom (edf) close to the maximum k' indicate that the basis may be too restrictive. For $k = 10$, the temperature smooth uses nearly all available degrees of freedom (edf = 8.68 out of $k' = 9$) and fails the diagnostic at the 5% level, suggesting that this specification is borderline constrained. With $k = 20$, the smooths use fewer degrees of freedom relative to the maximum (edf = 16.6 for temperature and 13.2 for precipitation), indicating greater flexibility and less binding constraints. Information criteria favor the GAMs over the quadratic model, with AIC selecting $k = 20$ and BIC slightly favoring the more parsimonious $k = 10$. Taken together, the diagnostic checks and fit criteria support using the $k = 20$ GAM as the preferred specification (Table 1).

Table 1: Model comparison and GAM basis-dimension diagnostics.

Model	AIC	BIC	df	edf $s(T)$	edf $s(P)$	k -check $s(T)$	k -check $s(P)$
GAM FE ($k = 20$)	5084.7	7256.0	300.0	16.6	13.2	0.97 (0.02)	0.98 (0.04)
GAM FE ($k = 10$)	5163.8	7236.0	286.3	8.68	8.27	0.98 (0.035)	0.99 (0.370)
FE quadratic	5429.0	7404.9	273.0	–	–	–	–

5.3 Estimated climate–GDP relationships

Figure 5 shows the estimated smooth functions for temperature and precipitation. The temperature response is strongly nonlinear: output rises at low to moderate temperatures, peaks at intermediate levels, and declines sharply at higher temperatures, producing an inverted-U shape. By contrast, the precipitation response is much flatter, with smaller and more gradual marginal effects across the observed range.

These patterns indicate that temperature is the dominant nonlinear driver of aggregate climate damages in the historical data, while precipitation plays a secondary and more muted role.

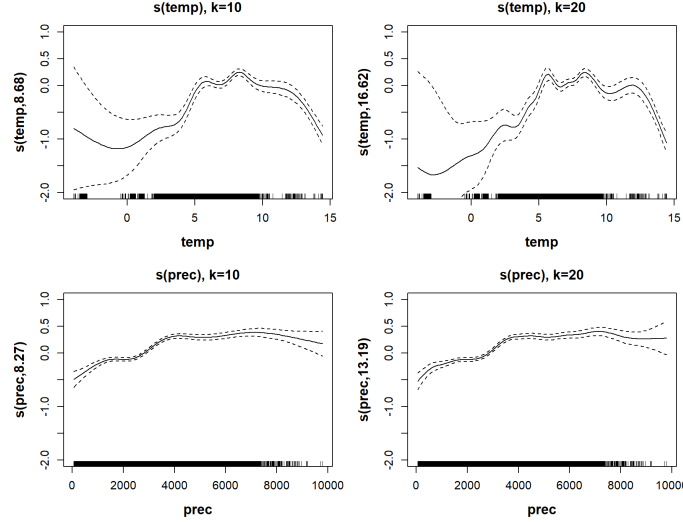


Figure 5: Estimated smooth functions for temperature and precipitation from GAM models with country and year fixed effects, comparing $k = 10$ and $k = 20$. Shaded bands denote approximate 95% confidence intervals.

6 Projections of Future Climate Change Impacts on GDP

6.1 Projection methodology

I use the preferred historical specification (the GAM with country and year fixed effects and basis dimension $k = 20$) to project climate-driven changes in GDP under alternative future scenarios. To isolate climate effects, I adopt a partial-effect approach: only the estimated smooth functions for temperature and precipitation are used for prediction, while country and year fixed effects are excluded. Projections therefore reflect climate variation alone, not assumptions about future institutions, technology, or global trends.

Let $\hat{f}_1(\cdot)$ and $\hat{f}_2(\cdot)$ denote the estimated smooth functions. For each country and year, the climate-only change in log GDP relative to a baseline climate (the country-specific average over 2011–2020) is

$$\Delta \log(\text{GDP})_{it}^{\text{climate}} = \left[\hat{f}_1(T_{it}) - \hat{f}_1(T_i^{\text{base}}) \right] + \left[\hat{f}_2(P_{it}) - \hat{f}_2(P_i^{\text{base}}) \right],$$

which is converted to a percentage change via

$$\% \Delta \text{GDP}_{it} = 100 \times \left(e^{\Delta \log(\text{GDP})_{it}^{\text{climate}}} - 1 \right).$$

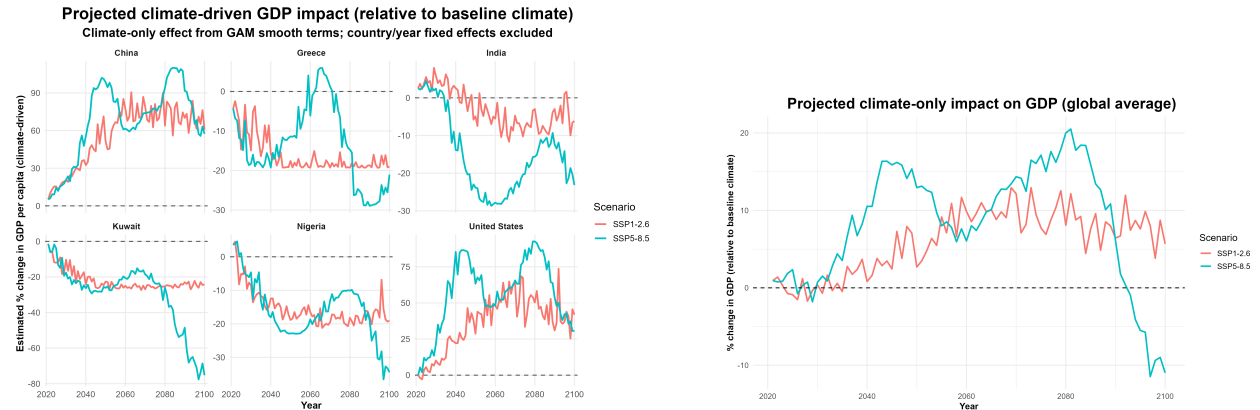
All projections are interpreted as *climate-only* impacts relative to a recent historical baseline.

6.2 Projected paths and cross-country heterogeneity

Figure 6a shows projected climate-driven percentage changes in GDP per capita from 2021 to 2100 under both scenarios. Impacts vary substantially across countries. Under the high-emissions scenario (SSP5-8.5), several countries experience increasingly negative effects late in the century, while others show non-monotonic paths reflecting nonlinear climate responses. Countries closer to historically productive temperature ranges (e.g., the United States or China) exhibit milder early impacts before turning negative, whereas hotter countries

(e.g., Kuwait or Iran) experience earlier and larger losses. Under the low-emissions scenario (SSP1-2.6), impacts are generally smaller and more stable.

Figure 6b reports the average climate-only impact across the six-country sample. The high-emissions scenario shows modest mid-century gains but becomes sharply negative by 2100, while the low-emissions scenario yields smaller and more stable impacts, highlighting the economic value of mitigation.



(a) Country-level projected climate-driven GDP impacts under SSP1-2.6 and SSP5-8.5.

(b) Average projected climate-only GDP impact across the six-country sample.

Figure 6: Projected climate-driven GDP impacts at the country level (left) and in the global average (right).

7 Interpretation and Policy Implications

- **Nonlinear temperature damages dominate:** The historical estimates show a clear inverted-U relationship between temperature and GDP (Figure 5), implying that economic damages accelerate once temperatures exceed historically productive ranges. Precipitation effects are flatter and smaller, indicating that temperature is the primary channel of aggregate climate-related economic risk.
- **Climate risk is inherently location-specific:** The projections imply that the same global warming path produces very different economic outcomes across countries, depending on initial climate conditions and position on the nonlinear damage curve. This reinforces the need for country-specific risk assessment rather than uniform global damage assumptions.
- **Downside risk dominates under high emissions:** The contrast between SSP1-2.6 and SSP5-8.5 shows that while some regions may experience temporary gains, long-run outcomes under high warming are strongly negative on average (Figure 6b). This asymmetry highlights the economic value of mitigation in limiting tail risks rather than just average losses.
- **Implications for climate insurance design:** Because damages rise nonlinearly with temperature, insurance triggers should be based on temperature thresholds and extreme heat exposure rather than linear deviations from historical means. Payout structures should reflect both nonlinear damage accumulation and country-specific exposure to these thresholds.

8 Recommendations for Strengthening the Assessment

- **Add sectoral detail:** Disaggregating GDP into agriculture, industry, and services would clarify which sectors drive aggregate climate sensitivity and improve the targeting of insurance instruments.
- **Incorporate adaptation and institutions:** Including measures of adaptive capacity (e.g., irrigation, infrastructure, governance) would help separate physical climate exposure from economic resilience.
- **Model climate extremes:** Supplementing annual averages with indicators of heat waves, droughts, or extreme precipitation would better capture tail risks most relevant for insurance design.
- **Quantify uncertainty and extrapolation risk:** Using climate-model ensembles and multiple scenarios, and propagating this uncertainty through the economic model, would yield probabilistic damage ranges and assess the reliability of out-of-sample projections.

9 Conclusion

This report analyzes historical climate–GDP relationships using a global panel from 1961–2020 and produces scenario-based projections for six countries from 2021–2100. The results show that temperature has a strong and highly nonlinear effect on economic output, while precipitation plays a more modest role. Projections based on these estimates imply large and heterogeneous climate-driven GDP impacts across countries, with substantially worse outcomes under the high-emissions pathway. Together, these findings highlight the importance of flexible nonlinear modeling, country-specific risk assessment, and provide a quantitative basis for designing climate insurance instruments that account for thresholds and tail risks.